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Question

While researching a topic, a student has taken the following notes:

- Gravitational waves are powerful ripples that originate in deep space and eventually pass through Earth.
- The Laser Interferometer Gravitational Wave Observatory (LIGO) is a physics study that began in 2002.
- LIGO's goal is to detect and analyze gravitational waves.
- LIGO uses a pair of massive gravitational wave detectors called interferometers that are thousands of miles apart.
- In 2015, for the first time in history, LIGO researchers detected a gravitational wave passing through Earth.

The student wants to present LIGO's aim and methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. In 2015, LIGO's massive interferometers detected a powerful ripple that originated in deep space and eventually passed through Earth.
- B. Though the physics study LIGO began in 2002, its massive interferometers didn't detect a gravitational wave until 2015.
- C. To achieve its aims, LIGO uses a pair of massive interferometers that are thousands of miles apart.
- D. A physics study designed to detect and analyze gravitational waves, LIGO uses a pair of massive interferometers that are thousands of miles apart.